R3. The Conversational Shopping Assistant That Doesn't Lie

Real-world pain: Retailers are rushing to put chatbots on product catalogues. Most of them confidently invent products, prices, and return policies. A hallucinated price is a legal and trust problem, not a cosmetic one.

The problem: Build a grounded product-discovery assistant over a fixed catalogue that answers natural shopper questions and **refuses to answer when the catalogue doesn't support it**.

What we provide:

- A catalogue of 5,000 products: title, description, attributes, price, stock status, category, ratings
- A policy document (returns, shipping, warranty) as plain text
- 150 test questions with hidden expected behaviour, deliberately including: answerable questions, ambiguous questions, out-of-catalogue questions ("do you sell iPhones?"), and **trap questions** ("what's the price of the blue running shoe?" when three exist)
- A small embedding model and small instruct model, pre-cached for offline use

Assumptions students may make:

- Catalogue is the single source of truth; nothing outside it may be asserted
- Prices and stock are current as of the snapshot
- No user account, order history, or payment integration exists
- Shoppers write short, ungrammatical, code-mixed queries

Desired output:

1. A working chat interface (notebook, CLI, or simple web page — all score equally)
2. **Every factual claim must carry a citation** to a product ID or policy section
3. An explicit refusal / clarification path — teams are scored on correctly saying "I don't have that" and on asking a disambiguating question rather than guessing
4. `evaluate.py` score = groundedness (no unsupported claims), answer helpfulness, and **refusal accuracy** — with hallucinated prices carrying a heavy penalty
5. A latency and token-cost report per query, since the budget is fixed and equal

Stretch: handle a multi-turn budget constraint ("under ₹2,000, and I already said I want cotton").